

Notes: Bordalo, Coffman, Gennaioli, and Shleifer (AER, 2019)
Beliefs about Gender

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1 Big picture

- **Question.** How do gender stereotypes shape beliefs about one’s own ability, beliefs about others’ ability, and behavior in group decision-making? The paper asks how much of observed gender gaps in confidence comes from stereotypes rather than task difficulty or generic overconfidence.
- **Setting.** The authors run laboratory experiments in which participants answer trivia questions across categories that differ in gender type: Kardashians, Disney, cooking, art, emotion recognition, verbal skills, business, math, rock and roll, sports and games, video games, and cars.
- **Core empirical challenge.** Beliefs can be distorted for multiple reasons. People may stereotype men and women differently across categories, but they may also overestimate performance on difficult questions. The paper’s central task is to separate these two forces.
- **Main mechanism 1: stereotypes.** The paper models stereotypes using the “kernel of truth” logic from Bordalo et al. (2016): stereotypes exaggerate true group differences. If men perform slightly better in a category, beliefs may overstate male advantage in that category.
- **Main mechanism 2: difficulty-induced misestimation (DIM).** Participants overestimate ability on hard questions and may underestimate ability on easy questions. DIM affects beliefs about both self and others, so it cannot be explained only by self-serving overconfidence.
- **Identification idea.** Stereotypes vary across categories according to the gender performance gap; DIM varies with question difficulty. Comparing easy and hard questions within a category helps identify DIM, while comparing equally difficult questions across gender-typed categories helps identify stereotypes.
- **Main belief result.** After controlling for DIM, stereotypes significantly shape self-beliefs and beliefs about others. The effects are strongest and most consistent for beliefs about women.
- **Main behavioral result.** Stereotypes also affect strategic choices in a cooperative game. Revealing the partner’s gender does not improve group performance; if anything, it lowers performance because stereotyped beliefs distort contribution decisions.
- **Economic takeaway.** Gender confidence gaps cannot be understood by looking only at average overconfidence. They reflect the interaction of category-specific stereotypes and generic difficulty-related belief distortions.

2 Incremental contribution of the paper

- **From documenting confidence gaps to decomposing belief distortions.** Prior work shows that men and women differ in confidence in some domains. This paper asks why: are the differences driven by gender stereotypes, by task difficulty, or by a broader tendency to overestimate performance?
- **A theory-guided measurement contribution.** The paper embeds a model of stereotypes into an empirical design. The key theoretical object is not a vague perception of gender type, but the exaggeration of true category-level gender performance gaps.
- **Separation of stereotypes from DIM.** The paper’s main methodological contribution is to separate category-level stereotypes from question-level difficulty-induced misestimation. This

matters because male-typed categories can also be harder for women, and difficulty alone would push beliefs in the opposite direction from stereotypes.

- **Beliefs about both self and others.** The paper does not study only self-confidence. It also elicits beliefs about partners' performance, showing that similar psychological forces shape self-assessments and assessments of others.
- **Use of partner-gender randomization.** By randomly varying whether a participant learns the partner's gender, the experiment can study how gender salience changes beliefs and behavior.
- **Behavioral link through a cooperative game.** The place-in-line game connects beliefs to actual choices over whether to contribute answers to the group. This allows the paper to show that stereotyped beliefs are not just survey responses; they affect group outcomes.
- **A unifying framework for confidence and discrimination.** The conclusion links confidence gaps and discrimination: both may reflect inaccurate beliefs produced by stereotypes and by difficulty-related misestimation.

3 Data and measurement

3.1 Experimental samples

- The paper reports three laboratory experiments conducted at Ohio State University, Harvard Business School/Harvard College, and the University of California Santa Barbara.
- The sample contains 548 men and 508 women after exclusions, with participants distributed across the three experimental sites.
- The experiments are based on a three-part design: participants first answer questions and assess own performance, then make contribution decisions in groups, and finally report detailed beliefs about own and partner performance.
- The UCSB experiment was added to include more strongly female-typed categories, so that the authors could estimate stereotypes using both male-advantage and female-advantage domains.

Interpretation: The sample is not a representative population sample. Its value is experimental control: the authors observe true ability, own beliefs, beliefs about others, partner gender information, and contribution decisions in the same environment.

3.2 Question categories

- The categories are chosen to span female-typed, male-typed, and ambiguous domains.
- Female-typed categories include Kardashians, Disney movies, cooking, art, emotion recognition, and verbal skills, though actual performance gaps are not always female-favoring.
- Male-typed categories include business, math, rock and roll, sports and games, video games, and cars.
- The paper also asks participants to rate which gender generally knows more about each category using a slider scale from female advantage to male advantage.

Interpretation: Category variation is essential. Stereotypes are defined by category-level beliefs about gendered ability, so a single math test would not allow the authors to separate stereotypes from generic task difficulty.

3.3 Own ability and self-beliefs

- In part 1, participants answer 10 multiple-choice questions per category.
- Participants report bank-level beliefs: their expected number correct in each 10-question category bank.
- In part 3, participants report question-level beliefs: their believed probability that their answer to a specific question is correct.
- The paper observes true performance and stated beliefs, allowing direct measurement of overestimation or underestimation.

Interpretation: The combination of bank-level and question-level beliefs is useful because the two measures identify difficulty-induced misestimation using different sources of variation.

3.4 Beliefs about others

- Participants are randomly assigned to two-person groups.
- In some treatments, the partner’s gender is revealed through a photograph or a voice response; in other treatments, the partner’s gender is not revealed.
- Participants report beliefs about the partner’s performance at both the bank level and the question level.
- For beliefs about others, the key observed object is whether the participant evaluates a known male or a known female partner differently after controlling for average ability and difficulty.

Interpretation: Beliefs about others are central because they connect the paper to discrimination and team allocation decisions, not only to self-confidence.

3.5 Place-in-line game

- In part 2, each participant answers new questions and chooses how willing they are to have their answer count for the group.
- For each question, participants choose a “place in line” from 1 to 4. A lower place in line means a greater willingness to contribute the answer.
- The group answer is determined by the participant with the lower place in line; ties are broken randomly.
- This game measures whether beliefs about relative ability translate into contribution decisions.

Interpretation: In a rational benchmark, learning a partner’s gender should help if gender predicts performance. The paper finds the opposite pattern, suggesting that stereotypes distort how gender information is used.

4 Models and empirical specifications

4.1 Ability decomposition

The paper begins with a simple decomposition of true ability. Let $p_{i,j}$ be the probability that individual i answers question j correctly. If individual i belongs to gender group G and question j belongs to category J , then

$$p_{i,j} = p_{G,J} + a_{i,j}.$$

- $p_{G,J}$ is the average performance of gender G in category J .
- $a_{i,j}$ captures individual-specific ability and question-specific difficulty relative to the gender-category average.
- The gender gap in a category is $p_{G,J} - p_{-G,J}$.

Interpretation: This decomposition separates average group-category ability from individual and question variation. It creates the structure needed to distinguish stereotypes from difficulty.

4.2 Difficulty-induced misestimation

The paper models difficulty-induced misestimation as an affine transformation of true ability:

$$p_{i,j}^{DIM} = c + \omega p_{i,j}.$$

- If $c > 0$ and $0 < \omega < 1$, beliefs are too high for hard questions where $p_{i,j}$ is low.
- The same specification can also imply mild underestimation for very easy questions.
- DIM is a reduced-form representation of several possible mechanisms: imperfect knowledge of difficulty, probability weighting, overprecision, bounded noisy beliefs, or other cognitive errors.

Interpretation: DIM is not a gender stereotype. It is a general belief distortion tied to difficulty. It matters because many gender-typed categories differ in average difficulty for men and women.

4.3 Stereotypes and kernel of truth

The stereotype component follows the kernel-of-truth model. When gender performance gaps are small, the stereotyped belief about the average ability of gender G in category J can be approximated by

$$p_{G,J}^{st} = p_{G,J} + \theta \sigma (p_{G,J} - p_{-G,J}).$$

- θ captures the strength of representativeness-driven distortion.
- σ captures how salient gender comparisons are in the situation.
- If $\theta \sigma > 0$, beliefs exaggerate the true gender gap.
- In a male-advantage category, the model predicts overestimation of male ability and underestimation of female ability.

Interpretation: The model does not assume stereotypes are completely false. Instead, stereotypes are biased because they overreact to real average differences.

4.4 Self-belief equation

Combining DIM and stereotypes gives the main question-level self-belief equation:

$$p_{i,j}^b = c + \omega(p_{G,J} + a_{i,j}) + \theta\sigma(p_{G,J} - p_{-G,J}).$$

- c and ω capture DIM.
- $\theta\sigma$ captures the stereotype response to the category-level gender performance gap.
- Rational expectations correspond to $c = 0$, $\omega = 1$, and $\theta\sigma = 0$.
- The bank-level version averages this equation over questions in a category.

4.5 Beliefs about others

For beliefs about a partner of known gender G , the corresponding question-level equation is

$$p_{i' \rightarrow i,j}^b = c + \omega(p_{G,J} + E_i[a_{i,j}]) + \theta\sigma(p_{G,J} - p_{-G,J}).$$

- The evaluator does not observe the partner's question-specific ability, so beliefs depend on the average hit rate of the partner's gender on that question.
- The same stereotype term applies because the evaluator uses gender-category information.
- Estimating this equation separately for beliefs about men and women reveals whether stereotypes operate symmetrically.

4.6 Measurement of ability in question-level regressions

For question-level self-beliefs, the authors do not observe the exact probability $p_{i,j}$; they observe only whether the participant got the question right. To reduce measurement error, they use a first-stage prediction:

$$I_{i,j} = \alpha_0 + \alpha_1 p_{i,J \setminus j} + \alpha_2 p_{G \cup -G \setminus i,j} + \alpha_3 (p_{G,J} - p_{-G,J}) + u_{i,j}.$$

- $p_{i,J \setminus j}$ is the individual's average performance in the same category excluding question j .
- $p_{G \cup -G \setminus i,j}$ is the average performance of other participants on the same question.
- The fitted value proxies for true individual-question ability.

Interpretation: This step is important because using the raw correct-answer dummy would create attenuation and potentially bias the estimated effect of stereotypes.

4.7 Behavioral specification

The contribution analysis connects beliefs to the place-in-line game. The implicit behavioral object is a participant's perceived relative ability:

$$BeliefGap_{i,J} = Belief(Self_{i,J}) - Belief(Partner_{i,J}).$$

- Participants should contribute more when they believe they are more likely than the partner to know the answer.
- If gender stereotypes distort relative beliefs, they should also distort contribution choices.
- The group-performance test compares groups where gender is known with groups where gender is unknown.

5 Identification and empirical design

5.1 What identifies stereotypes

- Stereotypes are identified from the relationship between beliefs and category-level gender gaps, after controlling for actual ability and difficulty.
- The identifying comparison is across categories: beliefs in male-advantage categories versus female-advantage categories, holding own or partner ability fixed.
- The kernel-of-truth model predicts that beliefs should move in the direction of the true gender gap but with exaggeration.

Interpretation: The empirical design treats the gender gap in actual performance as the object around which stereotypes are formed. This makes the stereotype measure disciplined and testable.

5.2 What identifies DIM

- DIM is identified from variation in difficulty across questions and across individuals.
- At the question level, the paper compares beliefs for hard and easy questions within and across categories.
- At the bank level, DIM is estimated from variation in average ability across individuals and category banks.

Interpretation: The slope $\omega < 1$ and intercept $c > 0$ are the empirical signature of difficulty-related misestimation: beliefs are compressed toward the middle and inflated for hard questions.

5.3 Role of orthogonality assumption

- The model assumes that DIM depends on task difficulty, whereas stereotypes depend on category-level gender gaps.
- This orthogonality is an approximation, not a law of nature.
- The authors use robustness checks with slider-scale perceptions and with strongly gender-typed categories to assess whether the stereotype estimates are sensitive to noisy measured gaps.

Interpretation: The paper’s design is strongest as a structured decomposition of belief distortions. It is not a field causal estimate of the full effect of gender norms on labor-market outcomes.

5.4 Randomization of partner gender information

- Partner gender revelation is experimentally varied.
- In a rational model, learning a partner’s gender should improve decisions when gender predicts ability.
- The paper instead finds that gender revelation can make beliefs more stereotyped and reduce group performance.

Interpretation: This is the most causal part of the paper. Randomizing whether gender is known allows the authors to study how the salience and use of gender information affects beliefs and behavior.

5.5 Main limitations

- The laboratory setting may not capture all real-world stakes, feedback, and institutional features.
- The categories are deliberately selected and may not map one-to-one to workplace domains.
- True gender gaps in some categories are estimated with noise.
- The model identifies a reduced-form stereotype coefficient, not the full social process through which stereotypes form.

6 Tables and figures

6.1 Table 1 and Figure 1: sample and raw gender gaps

Read:

- Table 1 reports sample composition: 548 men and 508 women, with participants from OSU, Harvard, and UCSB.
- The table also shows some demographic imbalance. Men are more likely to have attended a US high school and more likely to be Caucasian; women are more likely to be East Asian.
- Figure 1 compares actual male advantage in performance with male advantage in self-beliefs across categories.
- The categories are ordered from most female-typed to most male-typed according to the slider-scale perception.
- The believed gender gap generally moves with the actual performance gap and is often directionally larger, especially in more male-typed categories such as sports, video games, and cars.

Economic message: The raw data already suggest the kernel-of-truth pattern: beliefs reflect true gender performance gaps but tend to exaggerate them. However, the raw figure is not enough, because task difficulty also distorts beliefs.

TABLE 1—SUMMARY STATISTICS

	Men	Women	<i>p</i> -value
Proportion OSU participants	0.39	0.34	0.08
Proportion Harvard participants	0.23	0.25	0.64
Proportion UCSB participants	0.37	0.41	0.18
Current student	0.996	0.996	0.93
Attended US high school	0.90	0.85	0.02
<i>Ethnicity</i>			
Caucasian	0.54	0.36	0.00
East Asian	0.19	0.32	0.00
Latino	0.11	0.12	0.69
Black or African American	0.06	0.07	0.39
Observations	548	508	

Notes: *p*-value is given for the null hypothesis of no difference between genders using a two-tailed test of proportions. Two participants at Ohio State dropped out when photographs were taken. One participant at Ohio State was caught cheating (looking up answers on the internet); she was dismissed. One participant at Ohio State was unable to complete the experiment due to a computer failure. All observations from these participants and their randomly-assigned partners are excluded from the analysis. At UCSB, we preregistered a restriction to only participants who self-reported attending high school in the United States and thus we exclude non-US-HS UCSB participants.

(a) Table 1: Summary statistics

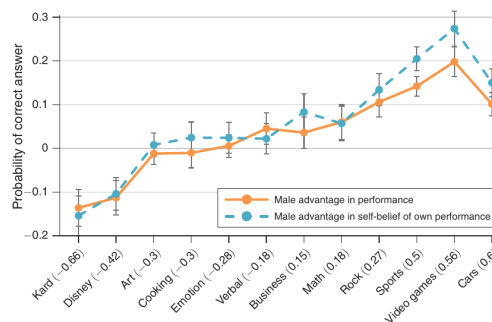


FIGURE 1. GENDER DIFFERENCES IN PERFORMANCE AND SELF-BELIEFS

Notes: Error bars reflect confidence intervals, where standard errors are clustered at the individual level. Average ider scale perceptions are in parentheses. We construct a measure of average ability in a category for each individual by calculating average probability of answering a question correctly across all 20 questions seen in the category. Then, we take the population average of this average ability measure by gender and difference the male and female averages. We construct a measure of average believed own ability in a category for each individual by first computing the average question-specific belief for that category for the individual (averaging over all questions in that category in part 3 answered by the individual) and then computing the average bank-level belief for that category for the

(b) Figure 1: Gender differences in performance and self-beliefs

6.2 Figure 2 and Figure 3: average beliefs and difficulty-induced misestimation

Read:

- Figure 2 compares average ability, own beliefs, and others' beliefs separately for men and women.
- Beliefs about both self and others often exceed actual ability, implying that overestimation is not only self-serving.
- Figure 3 plots question-level self-beliefs against actual question-level accuracy.
- Most points lie above the 45-degree line for hard questions, showing overestimation when the true probability of a correct answer is low.
- The fitted lines are flatter than the 45-degree line, consistent with the affine DIM specification $p^{DIM} = c + \omega p$ with $c > 0$ and $\omega < 1$.

Economic message: These figures motivate the need for the model. If difficult questions mechanically generate overestimation, then a simple comparison of confidence across gender-typed categories can understate or mask stereotypes.

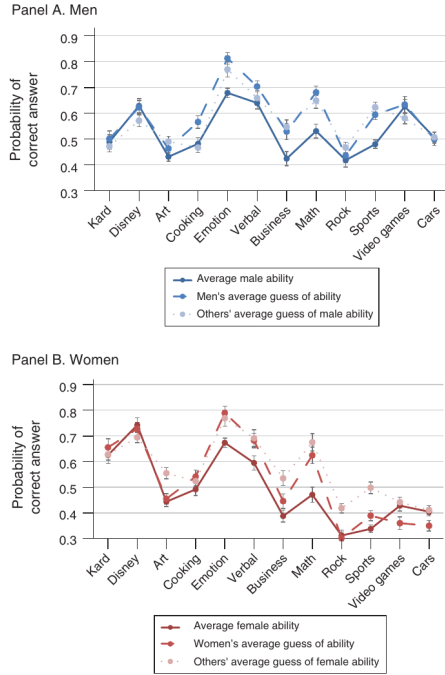


FIGURE 2. AVERAGE ABILITY AND BELIEFS

(a) Figure 2: Average ability and beliefs

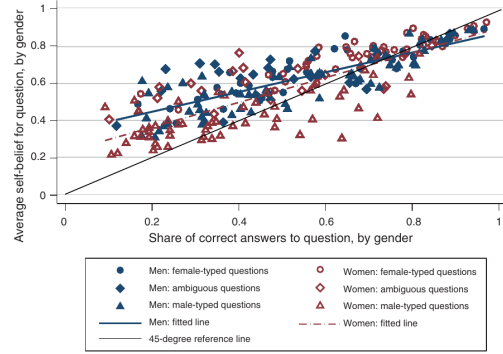


FIGURE 3. DIFFICULTY AND SELF-BELIEFS

Notes: Each point represents a question from the dataset, marking the averaged self-belief provided for that question

(b) Figure 3: Difficulty and self-beliefs

6.3 Table 2: gender gaps in performance

Read:

- Table 2 reports male advantage in the probability of a correct answer by category and question bank.
- Female performance exceeds male performance in Kardashians and Disney movies, with average gaps of about -13.7 and -11.3 percentage points.
- Male performance exceeds female performance in video games, sports and games, rock and roll, cars, math, verbal skills, and business.
- Several nominally female-typed categories, such as art, cooking, and emotion recognition, have small and statistically insignificant performance gaps.
- These measured gaps are the empirical input into the stereotype term.

Economic message: The table supplies the “kernel” in the kernel-of-truth model. Stereotypes are not identified from labels alone; they are tied to the observed gender performance gaps in each category.

6.4 Table 3 and Table 4: self-beliefs and beliefs about others

Read:

TABLE 2—SUMMARY STATISTICS ON GENDER GAPS IN PERFORMANCE

	Male advantage in prob. of correct answer at question-level			
	Average gap on bank 1 (M – W) (1)	Average gap on bank 2 (M – W) (2)	Average gap on both banks (M – W) (3)	<i>p</i> -value (Avg. gap on both banks = 0) (4)
Kardashians	–0.105 (0.021)	–0.169 (0.024)	–0.137 (0.021)	<0.001
Disney movies	–0.142 (0.022)	–0.084 (0.021)	–0.113 (0.020)	<0.001
Art	0.002 (0.016)	–0.026 (0.014)	–0.012 (0.013)	0.33
Cooking	0.002 (0.020)	–0.023 (0.019)	–0.010 (0.017)	0.55
Emotion recognition	0.024 (0.016)	–0.013 (0.017)	0.006 (0.013)	0.69
Business	0.079 (0.022)	–0.007 (0.020)	0.036 (0.018)	0.05
Verbal skills	0.062 (0.020)	0.029 (0.020)	0.045 (0.018)	0.01
Math	0.075 (0.022)	0.045 (0.022)	0.060 (0.020)	0.003
Cars	0.099 (0.017)	0.103 (0.015)	0.101 (0.013)	<0.001
Rock and roll	0.087 (0.019)	0.127 (0.020)	0.107 (0.017)	<0.001
Sports and games	0.142 (0.012)	0.142 (0.014)	0.142 (0.011)	<0.001
Video games	0.234 (0.021)	0.161 (0.018)	0.197 (0.017)	<0.001

Notes: Pools data from Ohio State, Harvard, and UCSB. Columns 2–4 report the mean difference in probability of answering correctly across gender in the ten-question bank. The standard error on the difference is reported in parentheses. *p*-value is given for the null hypothesis of no average performance difference between genders using a Fisher-Pitman permutation test for two independent samples.

Table 2: Summary statistics on gender gaps in performance

- Table 3 estimates the self-belief model using both question-level and bank-level beliefs.
- For women, stereotypes are strong and consistent: a category advantage for one’s own gender raises self-beliefs, and male advantage lowers women’s self-beliefs.
- For men, stereotypes are weaker and less consistent: the bank-level coefficient is positive, while the question-level coefficient is not significant in the predicted direction.
- DIM appears strongly in both male and female self-beliefs: constants are positive and slopes are below one.
- Table 4 repeats the analysis for beliefs about others among participants who know their partner’s gender.
- Beliefs about women are strongly affected by the partner’s gender advantage, especially in question-level beliefs. Beliefs about men are less consistently affected.

Economic message: The core empirical message is that stereotypes and DIM both matter. Stereotypes are not simply “men are overconfident”; the strongest and most stable effect is that women are evaluated, and evaluate themselves, less favorably in male-advantage domains.

TABLE 3—SELF-BELIEFS

Question-level self-beliefs: equation (5)			Bank-level self-beliefs: equation (6)		
Two-stage least squares predicting own believed probability of answering a question correctly			OLS predicting own believed score in bank on scale of 0 to 1		
Parameter	Men (1)	Women (2)	Parameter	Men (3)	Women (4)
Own gender advantage	$\theta\sigma$ (0.026)	0.49 (0.028)	Own gender advantage	$\theta\sigma$ (0.033)	0.44 (0.046)
Own ability: fitted value of $\hat{I}_{i,j}$	ω (0.011)	0.61 (0.011)	Own ability: own average probability of correct answer in bank	ω (0.018)	0.71 (0.020)
Constant	c (0.009)	0.33 (0.009)	Constant	c (0.012)	0.12 (0.012)
Clusters	548	504	Clusters	548	504
Observations	23,438	21,840	Observations	3,824	3,680

Notes: Pools observations for Ohio State, Harvard, and UCSB. Standard errors clustered at the individual level. *Own gender advantage* in both specifications is measured as the average gender difference in the probability of a correct answer on the bank of questions that the question is drawn from, coded so that a positive sign reflects an own gender advantage. *Own ability* for question-level data is the fitted value of $\hat{I}_{ij}$ from equation (9), and, in bank-level data, own ability is an individual's average probability of answering correctly in the bank. Bank-level beliefs and inputs are rescaled to a 0 to 1 scale; that is, while an individual predicts her score on a 0 to 10 scale, we divide that belief by 10 here, so that all coefficients can be interpreted in probability points.

(a) Table 3: Self-beliefs

TABLE 4—BELIEFS ABOUT OTHERS

Question-level beliefs: equation (7)			Bank-level beliefs: equation (8)				
OLS predicting belief of partner's probability of answering a question correctly			OLS predicting belief of partner's score on scale of 0 to 1				
	Parameter	Beliefs about men (1)	Beliefs about women (2)		Parameter	Beliefs about men (3)	Beliefs about women (4)
Partner's gender advantage	$\theta\sigma$	0.02 (0.027)	0.48 (0.037)	Partner's gender advantage	$\theta\sigma$	0.45 (0.052)	0.14 (0.055)
Partner ability: share of partner's gender answering question correctly	ω	0.34 (0.013)	0.33 (0.016)	Partner ability: partner's gender average probability of correct answer in bank	ω	0.64 (0.043)	0.62 (0.037)
Constant	c	0.40 (0.010)	0.43 (0.012)	Constant	c	0.16 (0.024)	0.21 (0.021)
Clusters		395	398	Clusters		395	398
Observations		18,020	18,179	Observations		2,590	2,630

Notes: Includes data only from participants who knew the gender of their partner at the time of providing the belief. Pools observations for Ohio State, Harvard, and UCSB. Standard errors clustered at the individual level. *Partner's gender advantage* in both specifications is measured as the average gender difference in the probability of a correct answer on the bank of questions that the question is drawn from, coded so that a positive sign reflects an advantage for the partner's gender. *Partner ability* for question-level data is the average probability of answering correctly in the 10 question bank by members of the partner's gender. Note that bank-level beliefs and inputs are rescaled to a 0 to 1 scale; that is, while an individual predicts her score on a 0 to 10 scale, we divide that belief by 10 here, so that all coefficients can be interpreted in probability points.

(b) Table 4: Beliefs about others

6.5 Table 5 and Table 6: counterfactual decomposition and context dependence

Read:

- Table 5 decomposes the male overconfidence gap into a DIM-only prediction, a stereotypes-only prediction, and the observed gap.
- DIM alone predicts more overconfidence for the gender that finds a category harder. This often works against the stereotype prediction.
- Stereotypes alone predict a stronger movement toward male overconfidence as categories become more male-typed.
- The observed pattern is muted relative to stereotypes alone because DIM is a countervailing force.
- Table 6 asks whether knowing the partner's gender changes self-beliefs by making gender comparisons more salient.
- The interaction patterns are directionally consistent with context dependence: beliefs become more stereotyped when the partner's gender makes gender comparison more salient, although individual coefficients are not always statistically significant.

Economic message: Table 5 is conceptually important because it shows why raw gender gaps in confidence can be misleading. DIM and stereotypes push in opposite directions in many domains. Table 6 adds that stereotypes are not fixed; they depend on context and salience.

	DIM-only prediction of gender gap in overconfidence (M – F) (1)	Stereotypes-only prediction of gender gap in overconfidence (M – F) (2)	Observed gender gap in overconfidence (M – F) (3)
Kardashians	0.078	–0.063	0.028
Disney	0.070	–0.046	0.024
Art	0.033	–0.018	0.021
Cooking	0.028	–0.006	0.042
Emotion	0.034	0.002	0.040
Verbal	0.016	0.019	–0.004
Business	0.028	–0.005	0.063
Math	0.008	0.022	0.022
Rock	–0.018	0.043	0.025
Sports	–0.032	0.062	0.038
Video games	–0.051	0.093	0.061
Cars	–0.015	0.045	0.033

Notes: We generate the DIM-only predictions by constructing a DIM-only question-level self-belief for each individual in the dataset. This uses the estimates presented in Table 3, but sets $\theta\sigma = 0$. Once we have generated this set of DIM-only beliefs for each individual, we use them to construct the counterfactual male overconfidence gap (MOG) by taking the average DIM-only self-belief for men (women), differencing out observed average male (female) ability, and then subtracting female overconfidence from male overconfidence. We follow the same approach for the stereotypes-only counterfactual, again using the estimates from Table 3, but this time setting

(a) Table 5: Predicted gender gaps in overconfidence

Question-level beliefs Two-stage least squares predicting own believed probability of answering a question correctly			Bank-level beliefs OLS predicting own believed score on 0 to 1 scale		
Parameter	Men (1)	Women (2)	Parameter	Men (3)	Women (4)
Own gender advantage $\theta\sigma$	–0.10 (0.036)	0.61 (0.038)	Own gender advantage $\theta\sigma$	0.11 (0.054)	0.51 (0.067)
Own ability— fitted value of $\hat{I}_{ij}$ ω	0.60 (0.013)	0.59 (0.012)	Own ability— own average probability of correct answer in bank ω	0.71 (0.021)	0.69 (0.022)
Partner female	–0.021 (0.012)	0.021 (0.012)	Partner Female	–0.014 (0.013)	0.000 (0.013)
Partner female $\times$ own gender advantage	0.045 (0.054)	–0.15 (0.056)	Partner female $\times$ own gender advantage	0.12 (0.076)	–0.068 (0.082)
Constant c	0.34 (0.013)	0.30 (0.010)	Constant c	0.12 (0.016)	0.11 (0.015)
Clusters	401	392	Clusters	401	392
Observations	18,359	17,840	Observations	2,612	2,608

Notes: Pools observations for Ohio State, Harvard, and UCSB. Standard errors clustered at the individual level. Uses only observations for individuals who knew their partner's gender at the time of the belief elicitation. *Own gender advantage* in both specifications is measured as the average gender difference in the probability of a correct answer on the bank of questions that the question is drawn from, coded so that a positive sign reflects an own gender advantage. *Own ability* for question-level data is the fitted value of $\hat{I}_{ij}$ from equation (9), and, in bank-level data, *own ability* is an individual's average probability of answering correctly in the bank. Note that bank-level beliefs are rescaled to a 0 to 1 scale: that is, while an individual predicts her score on a 0 to 10 scale, we divide that belief by 10 here so that all coefficients can be interpreted in probability points.

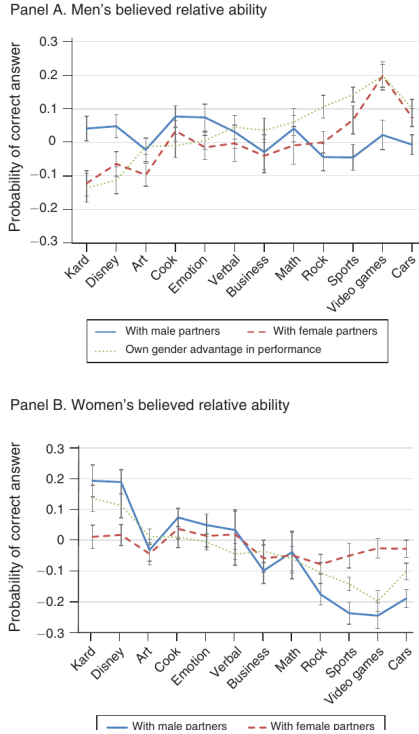
(b) Table 6: Self-beliefs with context dependence

6.6 Figure 4 and Figure 5: relative ability beliefs and group performance

Read:

- Figure 4 studies beliefs about relative ability: own believed ability minus partner's believed ability.
- Men paired with women believe they are less knowledgeable in female-typed categories and more knowledgeable in male-typed categories.
- Women paired with men believe they outperform in female-typed categories and are outperformed in male-typed categories.
- When paired with same-gender partners, relative beliefs are closer to zero.
- Figure 5 shows group performance in the place-in-line game.
- Groups in which gender is known perform slightly worse overall than groups in which gender is unknown, even though gender contains information about average ability.

Economic message: Gender information is useful only if it is used correctly. The experiment shows that revealing gender can trigger stereotyped relative beliefs, which distort contribution choices enough to offset any informational benefit.



(a) Figure 4: Believed relative ability

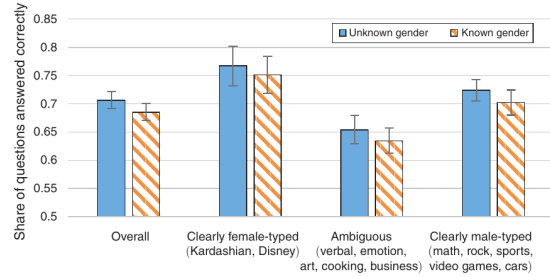


FIGURE 5. GROUP PERFORMANCE

†: Error bars reflect confidence intervals, where standard errors are clustered at the group level.

(b) Figure 5: Group performance

7 Short summary

- The paper studies how gender stereotypes shape beliefs about ability and behavior in laboratory experiments.
- Participants answer trivia questions across categories that differ in actual and perceived gender advantage.
- The paper separates two forces: category-level gender stereotypes and difficulty-induced miscalibration.
- Stereotypes follow a kernel-of-truth logic: beliefs exaggerate true category-level gender gaps.
- DIM means participants overestimate ability on difficult questions and sometimes underestimate ability on easy questions.
- Both stereotypes and DIM affect self-beliefs and beliefs about others.
- The stereotype effects are strongest and most consistent for women's self-beliefs and others' beliefs about women.
- DIM partly offsets stereotypes because people tend to be more overconfident in categories that are harder for them.
- Revealing a partner's gender can make beliefs more stereotyped and slightly reduce group performance.

- The paper links confidence gaps and discrimination by showing that beliefs about self and others are distorted by similar psychological forces.

8 Conclusion

8.1 Core conclusion

Bordalo, Coffman, Gennaioli, and Shleifer show that gendered beliefs are shaped by two distinct forces: stereotypes and difficulty-induced misestimation. Stereotypes exaggerate true gender performance gaps across domains, while DIM creates overestimation for difficult tasks. Because these forces can move in opposite directions, raw confidence gaps can understate the importance of stereotypes. The central empirical conclusion is that stereotypes meaningfully affect self-confidence, assessments of others, and behavior in a group decision task.

8.2 Economic intuition

The economic intuition is that people do not form beliefs only from their own performance signal. They also use category-level social knowledge about which gender is expected to be better in a domain. If that category-level information is over-weighted, a small average male advantage in math, sports, or video games can become a larger perceived male advantage. At the same time, hard questions produce generic overestimation for everyone. Thus, a woman in a hard male-typed domain faces two forces: difficulty pushes confidence upward, while stereotypes push confidence downward. The observed confidence gap is the net effect of both forces.

8.3 Incremental contribution revisited

The paper advances the gender-beliefs literature by converting the broad idea of stereotypes into an estimable model. It shows that stereotypes can be identified by comparing beliefs across categories with different gender performance gaps while separately controlling for difficulty. It also moves beyond self-confidence by studying beliefs about others and by linking those beliefs to actual contribution decisions. This makes the paper a bridge between behavioral models of stereotypes, experimental economics, and the economics of discrimination.

8.4 Implications for confidence and discrimination

The findings imply that interventions aimed at confidence gaps should distinguish between two mechanisms. If beliefs are distorted by DIM, then feedback about task difficulty and actual ability may help. If beliefs are distorted by stereotypes, then feedback alone may be insufficient unless it changes the perceived gender gap or reduces the salience of gender comparisons. The results also suggest that stereotypes in self-beliefs and stereotypes in evaluations of others are related phenomena. Reducing distorted beliefs about gender may therefore matter both for individual choices and for discriminatory allocation decisions by others.

8.5 Limitations

The evidence comes from laboratory trivia tasks, so external validity to educational choices, hiring, promotion, and financial decisions requires caution. The model requires an orthogonality assumption between difficulty-related misestimation and stereotype effects. Some category-level gender gaps are measured imprecisely, although the authors conduct robustness checks using slider-scale perceptions and stronger gender-typed domains. Finally, the paper identifies belief distortions in a controlled environment, not the full equilibrium process through which social stereotypes arise and evolve.

8.6 Takeaway

The enduring takeaway is that gender beliefs are neither purely accurate statistical inference nor simply generic overconfidence. They combine real performance differences, exaggerated stereotypes, and difficulty-related belief errors. The paper’s main contribution is to show how these forces can be separated empirically, and why doing so matters for understanding confidence, evaluation, and group performance.